

Sampling and Variability

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Tokens and Context

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute the adjusted table for Sampling and temperature. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sampling and temperature. Proposed explanation: “Sampling means equal chances”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Sample a new table independently

Probabilities $A=0.5, B=0.3, C=0.2$ in that order. Compute $T=0.5$ probabilities and cumulative boundaries. Trace $u=0.5$ and $u=0.9$ under $T=1$ and $T=0.5$. Explain the limit of the comparison.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

6. Solve the given case

One prompt produces three different valid quiz questions.

Does variability mean failure?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 4 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace boundaries

At $T=1$ select tokens for $u=0,0.6,0.9$ and 0.999 . Explain the handling of $u=1$.

My reasoning and evidence

13. W2 Compute the adjusted table

At $T=0.5$ compute squared weights, total and normalised probabilities. Check the sum.

My reasoning and evidence

14. W3 Compare a fixed draw

Trace $u=0.7$ at $T=1$ and $T=0.5$. Does the different output demonstrate improved factual accuracy?

My reasoning and evidence

15. W4 Sample a new table independently

Probabilities $A=0.5, B=0.3, C=0.2$ in that order. Compute $T=0.5$ probabilities and cumulative boundaries. Trace $u=0.5$ and $u=0.9$ under $T=1$ and $T=0.5$. Explain the limit of the comparison.

My reasoning and evidence

16. W5 Specify a valid experiment

A student changes the prompt and model as well as temperature, then attributes a better answer to temperature. Propose a fair comparison and reject any unsupported universal claim.

My reasoning and evidence

17. A small attention calculation — W1 Compute scores and weights

For $q=[1]$, keys $[0],[\ln(3)]$ and $d=1$, compute scores and softmax weights.

My reasoning and evidence

18. A small attention calculation — W2 Combine and mask

Use values [2],[6] for the preceding weights. Compute output and output when the second key is excluded.

My reasoning and evidence

19. A small attention calculation — W3 Trace equal scores

For $q=[1,1]$, keys $[1,0],[0,1]$, values $[2,0],[0,4]$, compute scaled scores, weights and output.

My reasoning and evidence

20. A small attention calculation — W4 Calculate new attention independently

$q=[1]$, keys $[0],[\ln(2)],[\ln(3)]$, values $[0],[3],[8]$, width 1. Compute scores, weights and output. Then mask the third item and recompute. Explain whether output establishes a factual claim.

My reasoning and evidence

21. A small attention calculation — W5 Check stability and shape

Scores are 1000 and $1000 + \ln(3)$; values 4 and 8. Compute stable shifted weights and output. What if one key has width two and the query width one?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Sampling and Variability

A fictional two-token distribution illustrates variability in generation.

At $T=1$, $A=0.8$ and $B=0.2$. For positive T use $p^{(1/T)}$, then normalise.

Use token order A,B and cumulative intervals $[0, \text{boundary})$, $[\text{boundary}, 1)$. Draw $u=0.85$.

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; $u=0.85$ selects B.



2. $T=0.5$ weights

Square probabilities: 0.64 and 0.04; total 0.68.



3. Normalise

$A=16/17 \approx 0.9412$; $B=1/17 \approx 0.0588$.



4. Same draw

0.85 is below $16/17$, so $T=0.5$ selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Keep Study Buddy generation settings fixed during prompt comparisons and check evidence regardless of variability.

24. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Keep Study Buddy generation settings fixed during prompt comparisons and check evidence regardless of variability.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.
